

NAME

ptpd - Precision Time Protocol daemon (1588-2008)

SYNOPSIS

```
ptpd [ -?hH ] [ -e SETTING ] [ -kvOLA1 ] [ -smMyEPanCV ] [ -c FILE ] [ -R DIR ] [ -f FILE ] [ -S FILE ] [ -d DOMAIN ] [ -u ADDRESS ] [ -r NUMBER ] -i INTERFACE
```

DESCRIPTION

PTPd is a daemon that implements the Precision Time Protocol (PTP) Version 2 as defined by the IEEE 1588-2008 standard. PTP was developed to provide very precise time coordination of LAN connected computers. The daemon must run as **root** in order to be able to manipulate the system clock and use low port numbers. PTPd is feature rich, supports IPv4 multicast, unicast and hybrid mode (mixed) operation, as well as Ethernet mode. Even without hardware assistance, PTPd is able to achieve and maintain sub-microsecond level timing precision and is able to withstand PTP Grandmaster failovers, link failures and restarts with minimal impact to timing performance. PTPd is lightweight, portable and currently supports Linux, FreeBSD and Mac OS X and runs on multiple CPU architectures, 32-bit and 64-bit, including x86 and ARM.

COMMAND-LINE CONFIGURATION

As of version 2.3.0, configuration file is the preferred mechanism for configuring PTPd, therefore the options available as short (*-x*) and long options (*--xxxxx*) mostly provide basic control over the daemon operation, and only provide the very basic PTP protocol settings. The rest of the settings (see [ptpd.conf\(5\)](#)) can also be specified as command-line options, but they take the long *--key:section="value"* form.

BASIC DAEMON OPTIONS

- c --config-file *PATH***
Path to configuration file (see [ptpd2.conf\(5\)](#))
- k --check-config**
Check configuration and exit - return 0 if configuration is correct.
- v --version**
Print version string and exit
- h --help**
Show help screen
- H --long-help**
Show detailed help for all settings and behaviours
- e --explain *SETTING***
Show help for a single setting (*section:key*)
- O --default-config**
Show default configuration and exit (output usable as a configuration file)
- L --ignore-lock**
Skip lock file checks and locking (also *global:ignore_lock*)
- A --auto-lock**
Use preset / port mode specific lock file names - useful when running multiple instances
- l --lockfile**

Specify lock file path (also *global:lock_file*)

-p --print-lockfile

Print path to lock file and exit (useful for init scripts in combination with auto lock files)

-R --lock-directory DIR

Directory to store lock files (also *global:lock_directory*)

-f --log-file PATH

Path to log file (also *global:logfile*)

-S --statistics-file I PATH

Path to statistics file (also *global:statistics_file*)

BASIC PTP PROTOCOL OPTIONS

-i --interface EV

*REQUIRED:*Interface to use - eth0, etc (also *ptpengine:interface*)

-d --domain NUMBER

PTP domain number to become part of (also *ptpengine:domain*)

-s --slaveonly

Slave only mode (also *ptpengine:preset=slaveonly*)

-m --masterslave

Full IEEE 1588 implementation: master, slave when not best GM (also *ptpengine:preset=masterslave*)

-M --masteronly

Master only mode: passive when not best GM (also *ptpengine:preset=masteronly*)

-y --hybrid

Hybrid mode - mixed multicast and unicast operation (multicast for sync and announce, unicast for delay request and response (also *ptpengine:ip_mode=hybrid*)

-u --unicast IP

Unicast mode (no unicast negotiation) - send all messages to *IP* (also *ptpengine:ip_mode=unicast* + *ptpengine:unicast_address*) **-E --e2e** End to end delay detection (also *ptpengine:delay_mechanism=E2E*)

-E --p2p Peer to peer delay detection (also *ptpengine:delay_mechanism=P2P*)

-a --delay-override

In slave state, override delay request interval announced by master (also *ptpengine:log_delayreq_override*) - the value of *ptpengine:log_delayreq_interval* is used

-r --delay-interval NUMBER

Specify delay request message interval (log 2) - (also *ptpengine:log_delayreq_interval*)

-n --clock:no_adjust

Do not adjust the clock (also *clock:no_adjust*)

-D<DD...> --debug

Debug level (also *global:debug_level*) - only if compiled with *RUNTIME_DEBUG*

-C --foreground

Don't run in background (also *global:foreground=Y*)

-V --verbose

Run in foreground, log all the messages to standard output (also *global:verbose_foreground=Y*)

COMPATIBILITY OPTIONS

PTPd supports the following options compatible with versions before 2.3.0:

-b <i>DEV</i>	Network interface to use
-i <i>NUMBER</i>	PTP domain number
-g	Slave only mode
-G	‘Master mode with NTP’ (master only mode)
-W	‘Master mode without NTP’ (master / slave mode)
-U	Hybrid mode (mixed unicast + multicast operation)
-Y <i>NUMBER</i>	Delay request interval (log 2)
-t	Do not adjust the clock

NOTE: the above options are deprecated and will be removed in subsequent versions. Until then, their use will issue a warning.

PTPD PORT STATES

<i>init</i>	INITIALIZING
<i>flt</i>	FAULTY
<i>lstn_init</i>	LISTENING (first time)
<i>lstn_reset</i>	LISTENING (subsequent reset)
<i>pass</i>	PASSIVE (not best master, not announcing)
<i>uncl</i>	UNCALIBRATED
<i>slv</i>	SLAVE
<i>pmst</i>	PRE-MASTER
<i>mst</i>	MASTER (active)
<i>dsbl</i>	DISABLED
<i>? (unk)</i>	UNKNOWN state

HANDLED SIGNALS

PTPd handles the following signals:

<i>SIGHUP</i>	Reload configuration file (if used) and reopen log files
<i>SIGUSR1</i>	When in slave state, force clock step to current Offset from Master value
<i>SIGUSR2</i>	Bahaviour based on <i>configure --enable-sigusr2=[value]</i> . Options: <i>domain</i> : swap domain between current and current +1; <i>debug</i> : cycle runtime debug level; <i>counters</i> : Dump all PTP protocol counters to current log target.
<i>SIGINT/SIGTERM</i>	

Clean exit - close logs and other open files, clean up lock file and exit.

SIGKILL Force an unclean exit.

EXIT CODES

Upon exit, `ptpd2` returns `0` on success - either successfully started in daemon mode, or otherwise exited cleanly. `0` is also returned when the `-k` (`--check-config`) option is used and the configuration was correct. A non-zero exit code is returned on errors. `3` is returned on lock file errors and when `ptpd2` could not be started as daemon. `2` is returned on memory allocation errors during startup. For all other error conditions such as configuration errors, running `ptpd2` in help mode or with no parameters, on self shutdown, network startup errors and when attempting to run `ptpd2` as non-root - `1` is returned.

SEE ALSO

`ptpd2.conf(5)`

AUTHORS

Gael Mace <gael_mace@users.sourceforge.net>

Alexandre Van Kempen

Steven Kreuzer <skreuzer@freebsd.org>

George Neville-Neil <gnn@freebsd.org>

Wojciech Owczarek <wojciech@owczarek.co.uk>

ptpd2(8) man page was written by Wojciech Owczarek for `ptpd 2.3.0` in November 2013
November, 2013 version 2.3.0